



MATH110: Normal Distribution

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Name: _____

Date: _____

Score: / 30

Learning Objectives

- Compute and interpret z-scores
- Apply the empirical rule (68-95-99.7)
- Relate raw scores to a standard normal distribution

Simplify each expression completely. Show all steps and circle your final answer.

Empirical rule (68-95-99.7)

1. Test scores are normally distributed with $\mu = 90$ and $\sigma = 4$. By the empirical rule, what percent of scores lie within 1 standard deviation(s) of the mean — that is, in the interval [86, 94]?

$$\mu = 90, \quad \sigma = 4, \quad [86, 94]$$

Answer: _____

2. Test scores are normally distributed with $\mu = 54$ and $\sigma = 8$. By the empirical rule, what percent of scores lie within 1 standard deviation(s) of the mean — that is, in the interval [46, 62]?

$$\mu = 54, \quad \sigma = 8, \quad [46, 62]$$

Answer: _____

3. Test scores are normally distributed with $\mu = 81$ and $\sigma = 6$. By the empirical rule, what percent of scores lie within 2 standard deviation(s) of the mean — that is, in the interval [69, 93]?

$$\mu = 81, \quad \sigma = 6, \quad [69, 93]$$

Answer: _____

4. Test scores are normally distributed with $\mu = 58$ and $\sigma = 8$. By the empirical rule, what percent of scores lie within 3 standard deviation(s) of the mean — that is, in the interval [34, 82]?

$$\mu = 58, \quad \sigma = 8, \quad [34, 82]$$

Answer: _____

5. Test scores are normally distributed with $\mu = 87$ and $\sigma = 5$. By the empirical rule, what percent of scores lie within 1 standard deviation(s) of the mean — that is, in the interval [82, 92]?

$$\mu = 87, \quad \sigma = 5, \quad [82, 92]$$

Answer: _____



6. Test scores are normally distributed with $\mu = 86$ and $\sigma = 5$. By the empirical rule, what percent of scores lie within 1 standard deviation(s) of the mean — that is, in the interval [81, 91]?

$$\mu = 86, \quad \sigma = 5, \quad [81, 91]$$

Answer: _____

7. Test scores are normally distributed with $\mu = 54$ and $\sigma = 6$. By the empirical rule, what percent of scores lie within 2 standard deviation(s) of the mean — that is, in the interval [42, 66]?

$$\mu = 54, \quad \sigma = 6, \quad [42, 66]$$

Answer: _____

8. Test scores are normally distributed with $\mu = 57$ and $\sigma = 6$. By the empirical rule, what percent of scores lie within 3 standard deviation(s) of the mean — that is, in the interval [39, 75]?

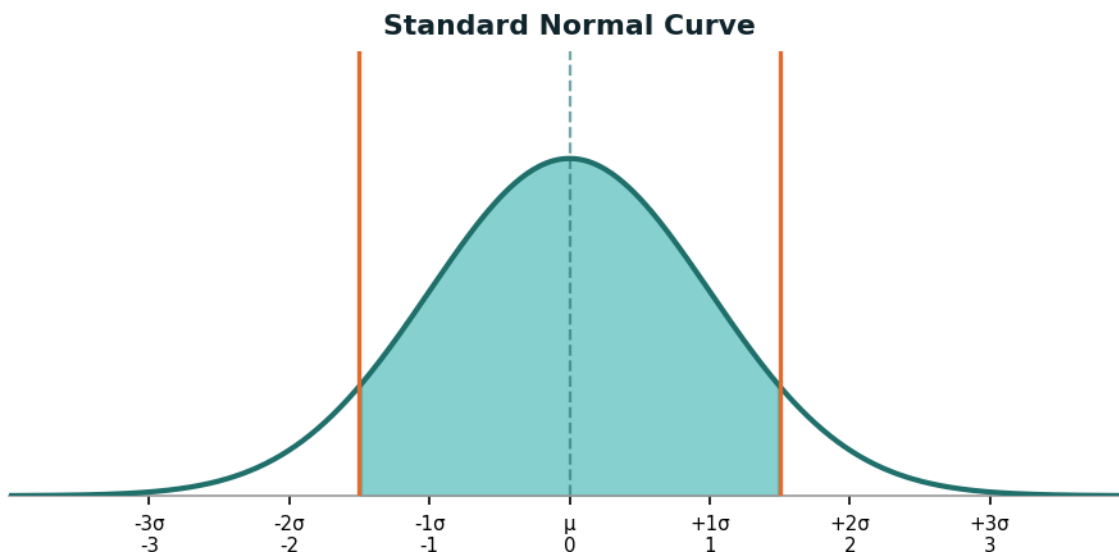
$$\mu = 57, \quad \sigma = 6, \quad [39, 75]$$

Answer: _____

Area Under the Normal Curve

9. Find the proportion of values in a standard normal distribution between $z = -1.5$ and $z = 1.5$. Use the z-table.

$$z_1 = -1.5, \quad z_2 = 1.5$$

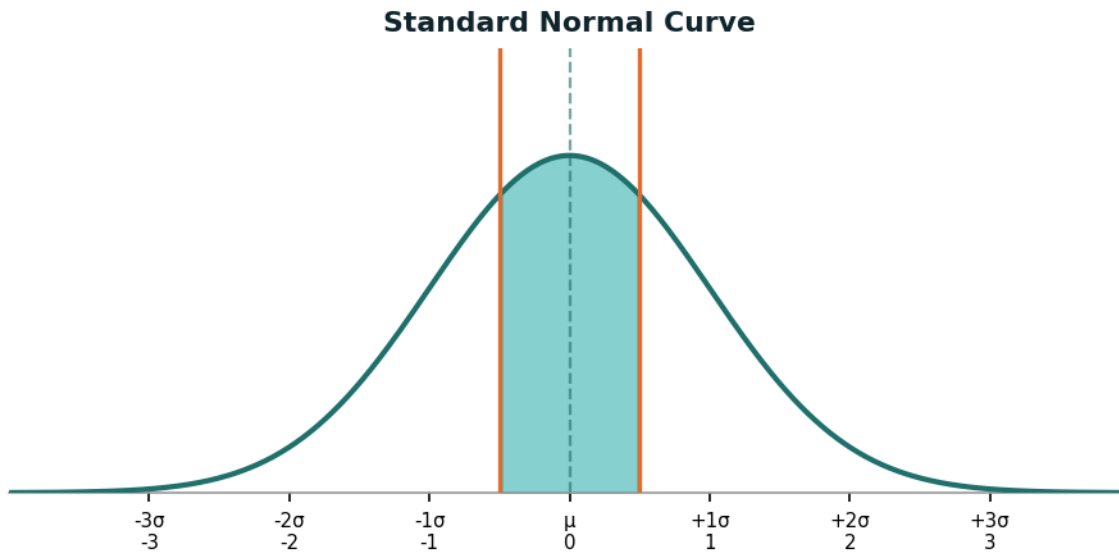


Answer: _____



10. Find the proportion of values in a standard normal distribution between $z = -0.5$ and $z = 0.5$. Use the z-table.

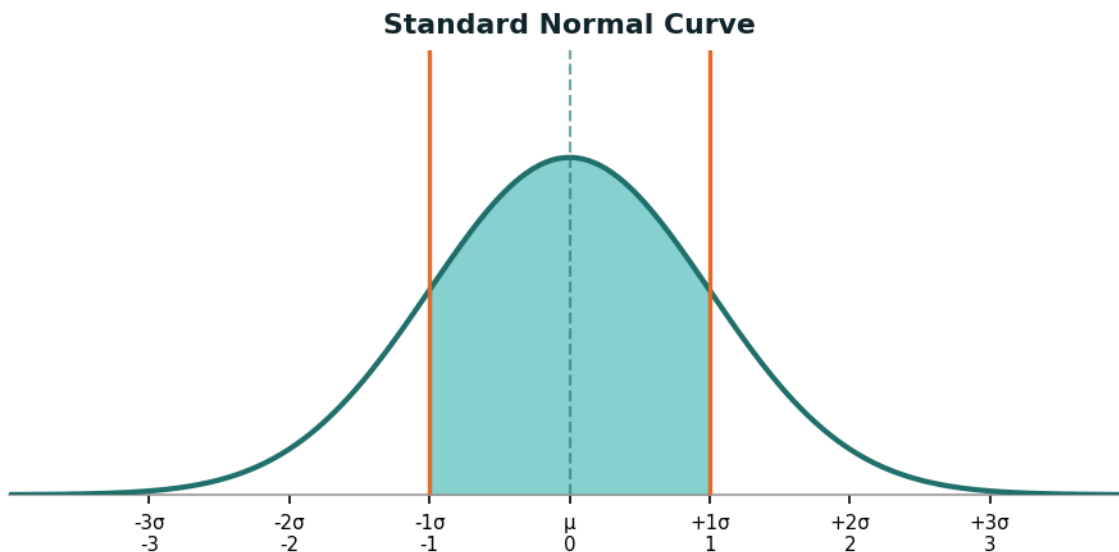
$$z_1 = -0.5, \quad z_2 = 0.5$$



Answer: _____

11. Find the proportion of values in a standard normal distribution between $z = -1.0$ and $z = 1.0$. Use the z-table.

$$z_1 = -1.0, \quad z_2 = 1.0$$

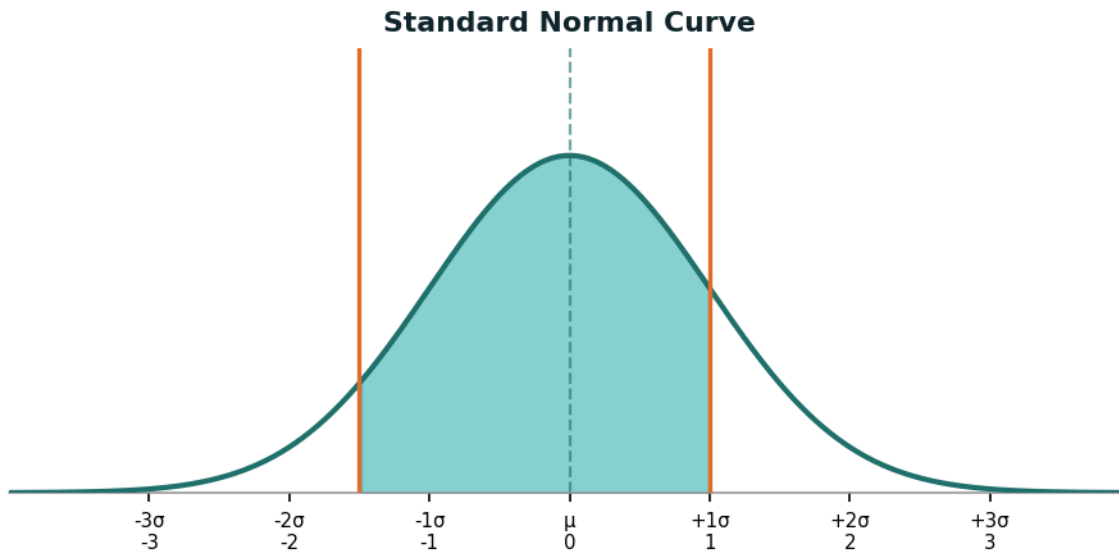


Answer: _____



12. Find the proportion of values in a standard normal distribution between $z = -1.5$ and $z = 1.0$. Use the z-table.

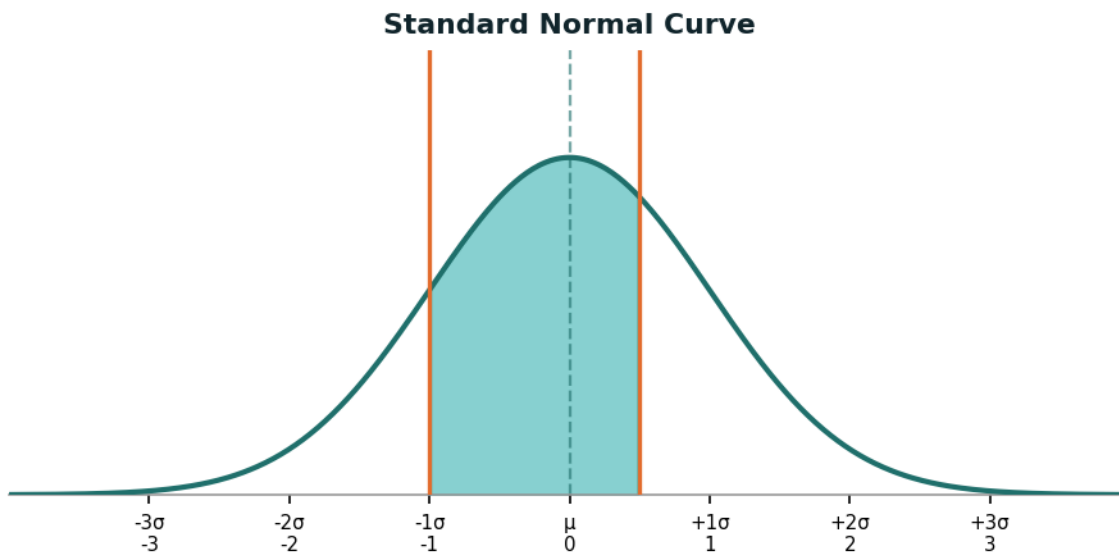
$$z_1 = -1.5, \quad z_2 = 1.0$$



Answer: _____

13. Find the proportion of values in a standard normal distribution between $z = -1.0$ and $z = 0.5$. Use the z-table.

$$z_1 = -1.0, \quad z_2 = 0.5$$

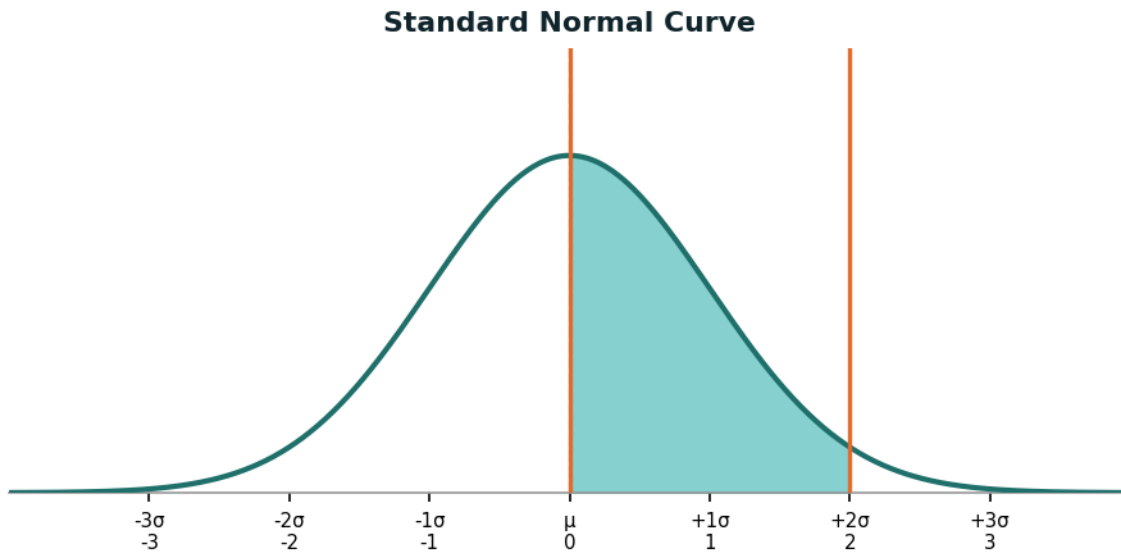


Answer: _____



14. Find the proportion of values in a standard normal distribution between $z = 0.0$ and $z = 2.0$. Use the z-table.

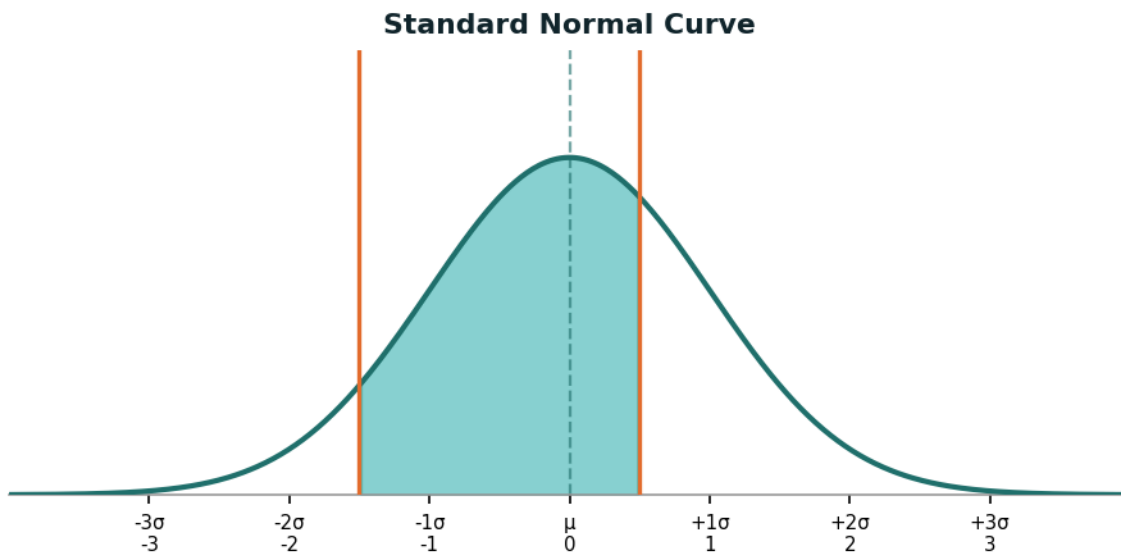
$$z_1 = 0.0, \quad z_2 = 2.0$$



Answer: _____

15. Find the proportion of values in a standard normal distribution between $z = -1.5$ and $z = 0.5$. Use the z-table.

$$z_1 = -1.5, \quad z_2 = 0.5$$

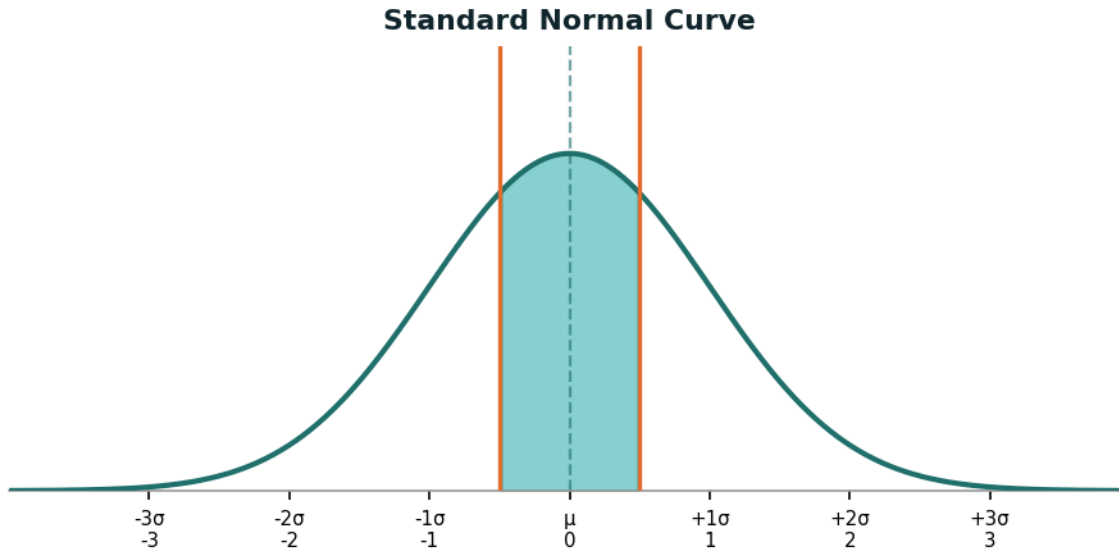


Answer: _____



16. Find the proportion of values in a standard normal distribution between $z = -0.5$ and $z = 0.5$. Use the z-table.

$$z_1 = -0.5, \quad z_2 = 0.5$$



Answer: _____

Finding a Score from a Percentile (Inverse Normal)

17. A distribution has mean $\mu = 76$ and standard deviation $\sigma = 8$. A score is at $z = -1.5$. Find the raw score X .

$$\mu = 76, \quad \sigma = 8, \quad z = -1.5$$

Answer: _____

18. A distribution has mean $\mu = 85$ and standard deviation $\sigma = 15$. A score is at $z = -1.5$. Find the raw score X .

$$\mu = 85, \quad \sigma = 15, \quad z = -1.5$$

Answer: _____

19. A distribution has mean $\mu = 67$ and standard deviation $\sigma = 6$. A score is at $z = 1.5$. Find the raw score X .

$$\mu = 67, \quad \sigma = 6, \quad z = 1.5$$

Answer: _____

20. A distribution has mean $\mu = 85$ and standard deviation $\sigma = 5$. A score is at $z = 0.5$. Find the raw score X .

$$\mu = 85, \quad \sigma = 5, \quad z = 0.5$$

Answer: _____



21. A distribution has mean $\mu = 69$ and standard deviation $\sigma = 14$. A score is at $z = 1.0$. Find the raw score X .

$$\mu = 69, \quad \sigma = 14, \quad z = 1.0$$

Answer: _____

22. A distribution has mean $\mu = 80$ and standard deviation $\sigma = 8$. A score is at $z = 1.5$. Find the raw score X .

$$\mu = 80, \quad \sigma = 8, \quad z = 1.5$$

Answer: _____

23. A distribution has mean $\mu = 97$ and standard deviation $\sigma = 19$. A score is at $z = 1.5$. Find the raw score X .

$$\mu = 97, \quad \sigma = 19, \quad z = 1.5$$

Answer: _____

Z-score from a normal distribution

24. A normal distribution has mean $\mu = 67$ and standard deviation $\sigma = 9$. Find the z-score for $x = 78$ and interpret it.

$$x = 78, \quad \mu = 67, \quad \sigma = 9$$

Answer: _____

25. A normal distribution has mean $\mu = 54$ and standard deviation $\sigma = 8$. Find the z-score for $x = 60$ and interpret it.

$$x = 60, \quad \mu = 54, \quad \sigma = 8$$

Answer: _____

26. A normal distribution has mean $\mu = 89$ and standard deviation $\sigma = 6$. Find the z-score for $x = 98$ and interpret it.

$$x = 98, \quad \mu = 89, \quad \sigma = 6$$

Answer: _____

27. A normal distribution has mean $\mu = 52$ and standard deviation $\sigma = 8$. Find the z-score for $x = 69$ and interpret it.

$$x = 69, \quad \mu = 52, \quad \sigma = 8$$

Answer: _____



28. A normal distribution has mean $\mu = 81$ and standard deviation $\sigma = 5$. Find the z-score for $x = 96$ and interpret it.

$$x = 96, \quad \mu = 81, \quad \sigma = 5$$

Answer: _____

29. A normal distribution has mean $\mu = 76$ and standard deviation $\sigma = 7$. Find the z-score for $x = 74$ and interpret it.

$$x = 74, \quad \mu = 76, \quad \sigma = 7$$

Answer: _____

30. A normal distribution has mean $\mu = 93$ and standard deviation $\sigma = 3$. Find the z-score for $x = 79$ and interpret it.

$$x = 79, \quad \mu = 93, \quad \sigma = 3$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Area Under the Normal Curve, Empirical rule (68-95-99.7), Finding a Score from a Percentile (Inverse Normal), Z-score from a normal distribution. All answers verified by independent computation.

Solutions



Empirical rule (68-95-99.7)

1. Test scores are normally distributed with $\mu = 90$ and $\sigma = 4$. By the empirical rule, what percent of scores lie within 1 standard deviation(s) of the mean — that is, in the interval [86, 94]?

$$\mu = 90, \quad \sigma = 4, \quad [86, 94]$$

→ Empirical rule: 68% within 1σ , 95% within 2σ , 99.7% within 3σ .

→ Within 1σ the interval is [86, 94] → 68%.

Answer: $86 = \mu - 1\sigma, \quad 94 = \mu + 1\sigma \Rightarrow 68\%$

2. Test scores are normally distributed with $\mu = 54$ and $\sigma = 8$. By the empirical rule, what percent of scores lie within 1 standard deviation(s) of the mean — that is, in the interval [46, 62]?

$$\mu = 54, \quad \sigma = 8, \quad [46, 62]$$

→ Empirical rule: 68% within 1σ , 95% within 2σ , 99.7% within 3σ .

→ Within 1σ the interval is [46, 62] → 68%.

Answer: $46 = \mu - 1\sigma, \quad 62 = \mu + 1\sigma \Rightarrow 68\%$

3. Test scores are normally distributed with $\mu = 81$ and $\sigma = 6$. By the empirical rule, what percent of scores lie within 2 standard deviation(s) of the mean — that is, in the interval [69, 93]?

$$\mu = 81, \quad \sigma = 6, \quad [69, 93]$$

→ Empirical rule: 68% within 1σ , 95% within 2σ , 99.7% within 3σ .

→ Within 2σ the interval is [69, 93] → 95%.

Answer: $69 = \mu - 2\sigma, \quad 93 = \mu + 2\sigma \Rightarrow 95\%$

4. Test scores are normally distributed with $\mu = 58$ and $\sigma = 8$. By the empirical rule, what percent of scores lie within 3 standard deviation(s) of the mean — that is, in the interval [34, 82]?

$$\mu = 58, \quad \sigma = 8, \quad [34, 82]$$

→ Empirical rule: 68% within 1σ , 95% within 2σ , 99.7% within 3σ .

→ Within 3σ the interval is [34, 82] → 99.7%.

Answer: $34 = \mu - 3\sigma, \quad 82 = \mu + 3\sigma \Rightarrow 99.7\%$

5. Test scores are normally distributed with $\mu = 87$ and $\sigma = 5$. By the empirical rule, what percent of scores lie within 1 standard deviation(s) of the mean — that is, in the interval [82, 92]?

$$\mu = 87, \quad \sigma = 5, \quad [82, 92]$$

→ Empirical rule: 68% within 1σ , 95% within 2σ , 99.7% within 3σ .

→ Within 1σ the interval is [82, 92] → 68%.

Answer: $82 = \mu - 1\sigma, \quad 92 = \mu + 1\sigma \Rightarrow 68\%$



6. Test scores are normally distributed with $\mu = 86$ and $\sigma = 5$. By the empirical rule, what percent of scores lie within 1 standard deviation(s) of the mean — that is, in the interval $[81, 91]$?

$$\mu = 86, \quad \sigma = 5, \quad [81, 91]$$

→ Empirical rule: 68% within 1σ , 95% within 2σ , 99.7% within 3σ .

→ Within 1σ the interval is $[81, 91] \rightarrow 68\%$.

Answer: $81 = \mu - 1\sigma, \quad 91 = \mu + 1\sigma \Rightarrow 68\%$

7. Test scores are normally distributed with $\mu = 54$ and $\sigma = 6$. By the empirical rule, what percent of scores lie within 2 standard deviation(s) of the mean — that is, in the interval $[42, 66]$?

$$\mu = 54, \quad \sigma = 6, \quad [42, 66]$$

→ Empirical rule: 68% within 1σ , 95% within 2σ , 99.7% within 3σ .

→ Within 2σ the interval is $[42, 66] \rightarrow 95\%$.

Answer: $42 = \mu - 2\sigma, \quad 66 = \mu + 2\sigma \Rightarrow 95\%$

8. Test scores are normally distributed with $\mu = 57$ and $\sigma = 6$. By the empirical rule, what percent of scores lie within 3 standard deviation(s) of the mean — that is, in the interval $[39, 75]$?

$$\mu = 57, \quad \sigma = 6, \quad [39, 75]$$

→ Empirical rule: 68% within 1σ , 95% within 2σ , 99.7% within 3σ .

→ Within 3σ the interval is $[39, 75] \rightarrow 99.7\%$.

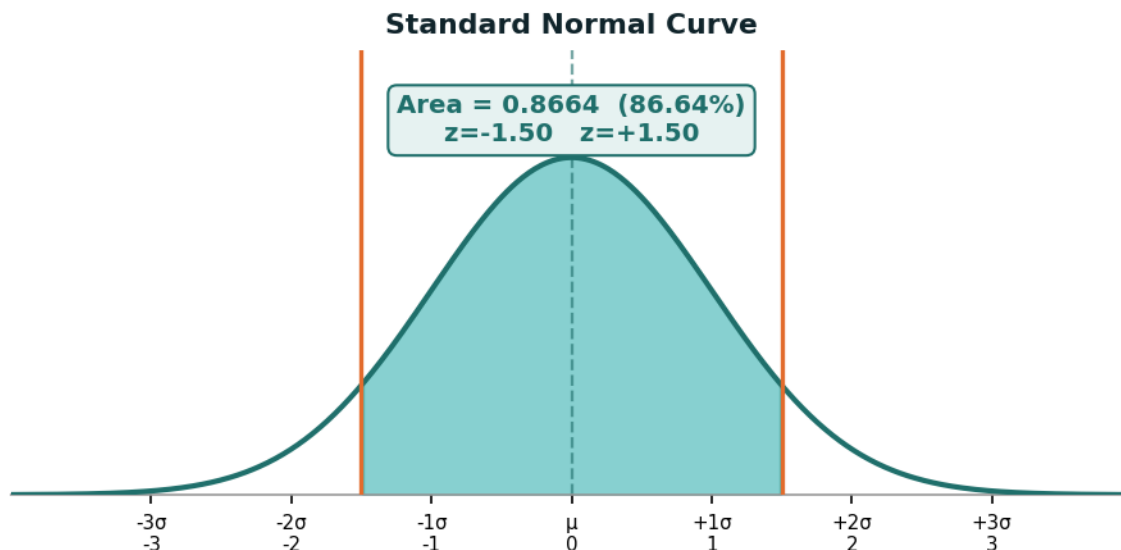
Answer: $39 = \mu - 3\sigma, \quad 75 = \mu + 3\sigma \Rightarrow 99.7\%$



Area Under the Normal Curve

9. Find the proportion of values in a standard normal distribution between $z = -1.5$ and $z = 1.5$. Use the z-table.

$$z_1 = -1.5, \quad z_2 = 1.5$$



→ Look up $\Phi(-1.5) = 0.0668$ (area to the left of $z = -1.5$).

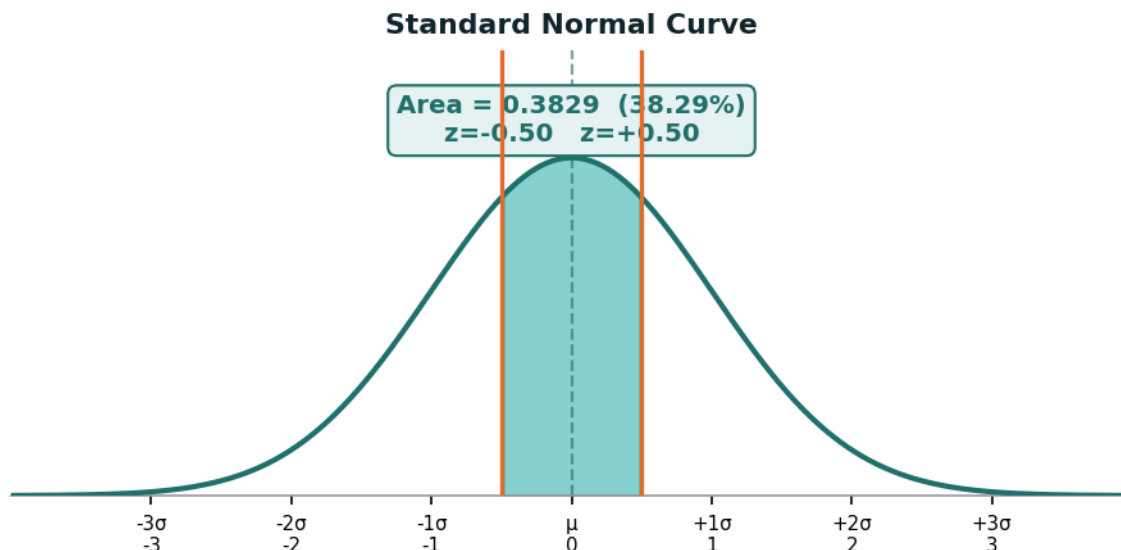
→ Look up $\Phi(1.5) = 0.9332$ (area to the left of $z = 1.5$).

→ Area between $z = -1.5$ and $z = 1.5 = 0.9332 - 0.0668 = 0.8664$.

Answer: $\Phi(1.5) - \Phi(-1.5) = 0.9332 - 0.0668 = 0.8664$

10. Find the proportion of values in a standard normal distribution between $z = -0.5$ and $z = 0.5$. Use the z-table.

$$z_1 = -0.5, \quad z_2 = 0.5$$



→ Look up $\Phi(-0.5) = 0.3085$ (area to the left of $z = -0.5$).

→ Look up $\Phi(0.5) = 0.6915$ (area to the left of $z = 0.5$).

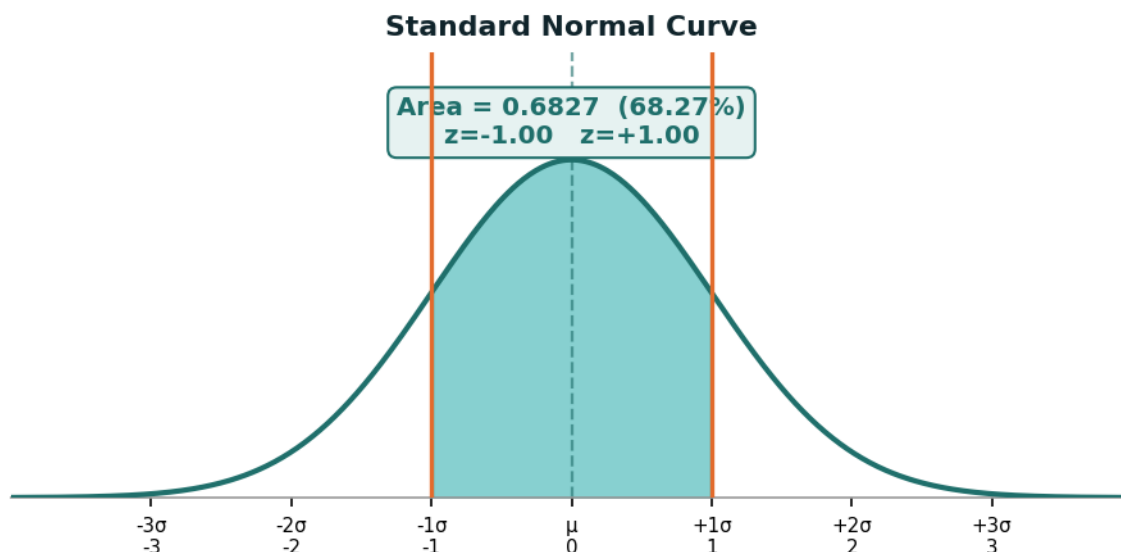
→ Area between $z = -0.5$ and $z = 0.5 = 0.6915 - 0.3085 = 0.3829$.

Answer: $\Phi(0.5) - \Phi(-0.5) = 0.6915 - 0.3085 = 0.3829$



11. Find the proportion of values in a standard normal distribution between $z = -1.0$ and $z = 1.0$. Use the z-table.

$$z_1 = -1.0, \quad z_2 = 1.0$$



→ Look up $\Phi(-1.0) = 0.1587$ (area to the left of $z = -1.0$).

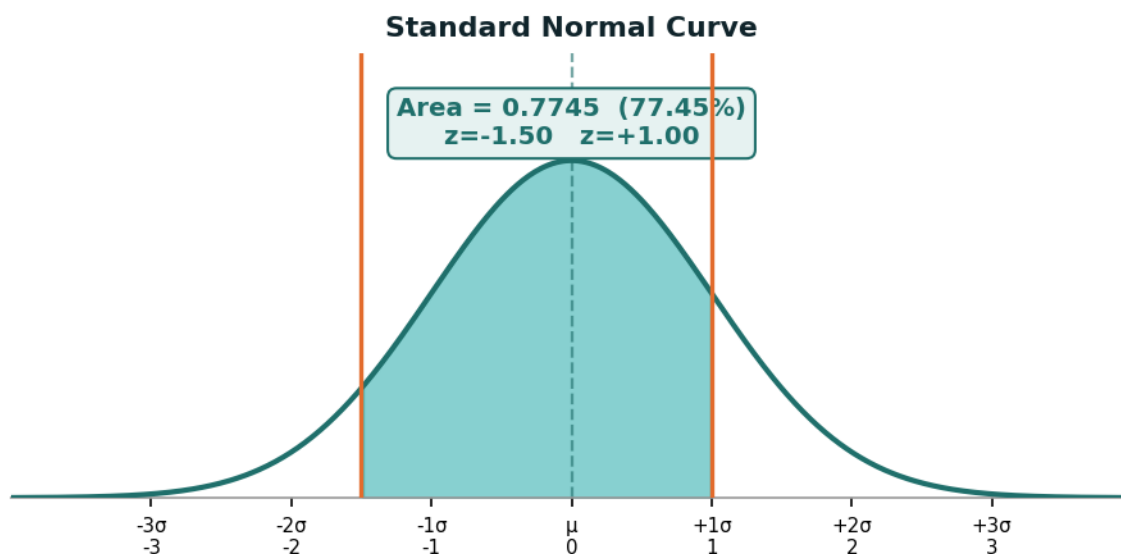
→ Look up $\Phi(1.0) = 0.8413$ (area to the left of $z = 1.0$).

→ Area between $z = -1.0$ and $z = 1.0 = 0.8413 - 0.1587 = 0.6827$.

Answer: $\Phi(1.0) - \Phi(-1.0) = 0.8413 - 0.1587 = 0.6827$

12. Find the proportion of values in a standard normal distribution between $z = -1.5$ and $z = 1.0$. Use the z-table.

$$z_1 = -1.5, \quad z_2 = 1.0$$



→ Look up $\Phi(-1.5) = 0.0668$ (area to the left of $z = -1.5$).

→ Look up $\Phi(1.0) = 0.8413$ (area to the left of $z = 1.0$).

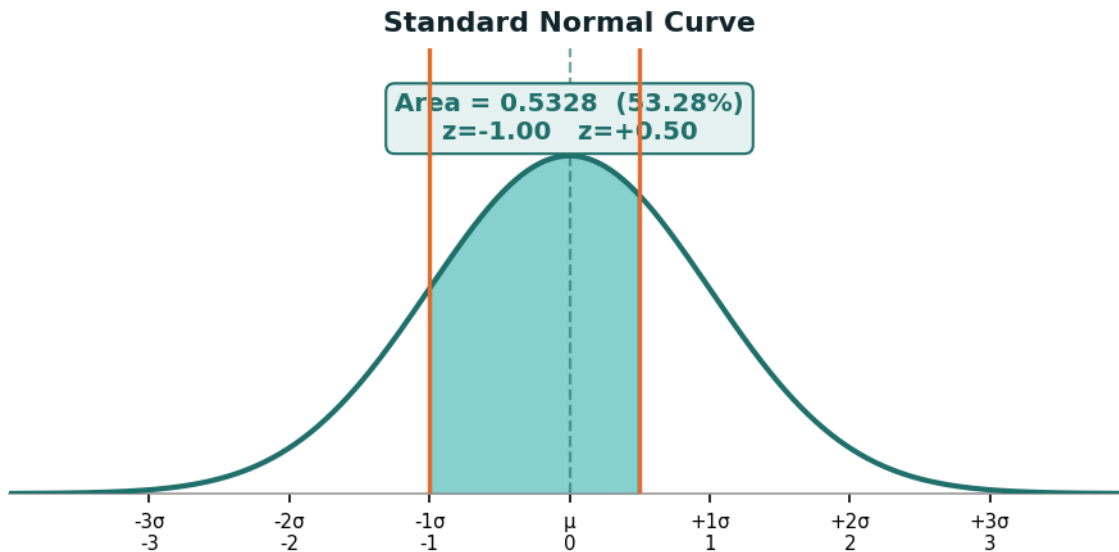
→ Area between $z = -1.5$ and $z = 1.0 = 0.8413 - 0.0668 = 0.7745$.

Answer: $\Phi(1.0) - \Phi(-1.5) = 0.8413 - 0.0668 = 0.7745$



13. Find the proportion of values in a standard normal distribution between $z = -1.0$ and $z = 0.5$. Use the z-table.

$$z_1 = -1.0, \quad z_2 = 0.5$$



→ Look up $\Phi(-1.0) = 0.1587$ (area to the left of $z = -1.0$).

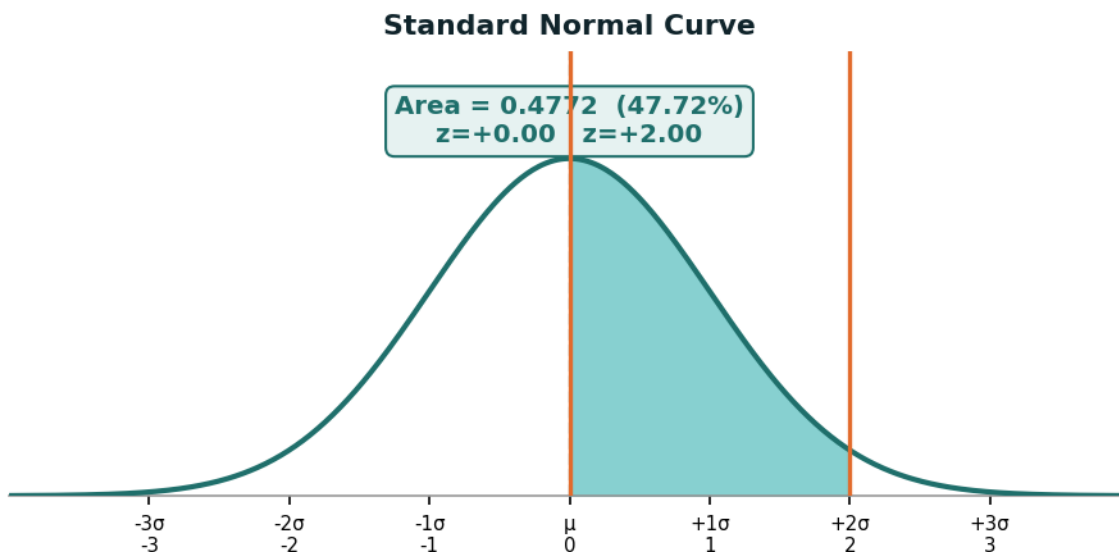
→ Look up $\Phi(0.5) = 0.6915$ (area to the left of $z = 0.5$).

→ Area between $z = -1.0$ and $z = 0.5 = 0.6915 - 0.1587 = 0.5328$.

Answer: $\Phi(0.5) - \Phi(-1.0) = 0.6915 - 0.1587 = 0.5328$

14. Find the proportion of values in a standard normal distribution between $z = 0.0$ and $z = 2.0$. Use the z-table.

$$z_1 = 0.0, \quad z_2 = 2.0$$



→ Look up $\Phi(0.0) = 0.5$ (area to the left of $z = 0.0$).

→ Look up $\Phi(2.0) = 0.9772$ (area to the left of $z = 2.0$).

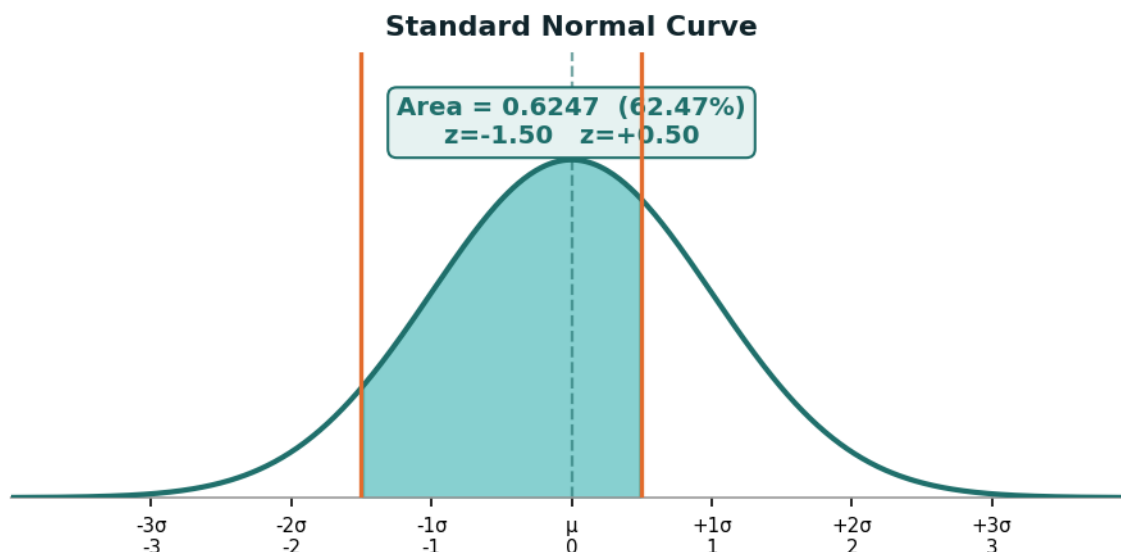
→ Area between $z = 0.0$ and $z = 2.0 = 0.9772 - 0.5 = 0.4772$.

Answer: $\Phi(2.0) - \Phi(0.0) = 0.9772 - 0.5 = 0.4772$



15. Find the proportion of values in a standard normal distribution between $z = -1.5$ and $z = 0.5$. Use the z-table.

$$z_1 = -1.5, \quad z_2 = 0.5$$



→ Look up $\Phi(-1.5) = 0.0668$ (area to the left of $z = -1.5$).

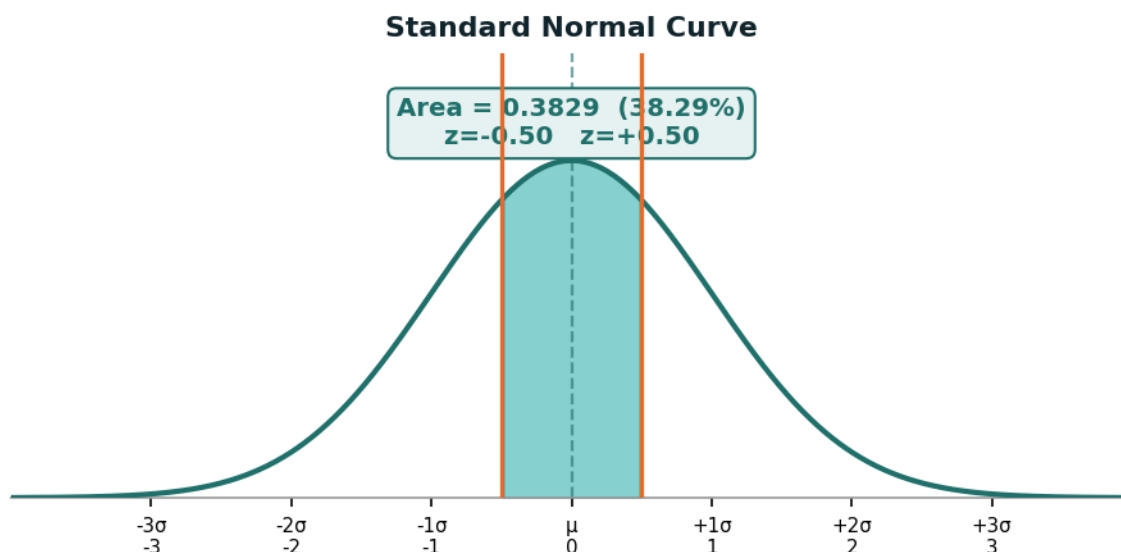
→ Look up $\Phi(0.5) = 0.6915$ (area to the left of $z = 0.5$).

→ Area between $z = -1.5$ and $z = 0.5 = 0.6915 - 0.0668 = 0.6247$.

Answer: $\Phi(0.5) - \Phi(-1.5) = 0.6915 - 0.0668 = 0.6247$

16. Find the proportion of values in a standard normal distribution between $z = -0.5$ and $z = 0.5$. Use the z-table.

$$z_1 = -0.5, \quad z_2 = 0.5$$



→ Look up $\Phi(-0.5) = 0.3085$ (area to the left of $z = -0.5$).

→ Look up $\Phi(0.5) = 0.6915$ (area to the left of $z = 0.5$).

→ Area between $z = -0.5$ and $z = 0.5 = 0.6915 - 0.3085 = 0.3829$.

Answer: $\Phi(0.5) - \Phi(-0.5) = 0.6915 - 0.3085 = 0.3829$



Finding a Score from a Percentile (Inverse Normal)

17. A distribution has mean $\mu = 76$ and standard deviation $\sigma = 8$. A score is at $z = -1.5$. Find the raw score X .

$$\mu = 76, \quad \sigma = 8, \quad z = -1.5$$

→ Use the formula: $X = \mu + z * \sigma$.

$$\rightarrow X = 76 + (-1.5)(8).$$

$$\rightarrow X = 64.0.$$

Answer: $X = 76 + (-1.5)(8) = 64.0$

18. A distribution has mean $\mu = 85$ and standard deviation $\sigma = 15$. A score is at $z = -1.5$. Find the raw score X .

$$\mu = 85, \quad \sigma = 15, \quad z = -1.5$$

→ Use the formula: $X = \mu + z * \sigma$.

$$\rightarrow X = 85 + (-1.5)(15).$$

$$\rightarrow X = 62.5.$$

Answer: $X = 85 + (-1.5)(15) = 62.5$

19. A distribution has mean $\mu = 67$ and standard deviation $\sigma = 6$. A score is at $z = 1.5$. Find the raw score X .

$$\mu = 67, \quad \sigma = 6, \quad z = 1.5$$

→ Use the formula: $X = \mu + z * \sigma$.

$$\rightarrow X = 67 + (1.5)(6).$$

$$\rightarrow X = 76.0.$$

Answer: $X = 67 + (1.5)(6) = 76.0$

20. A distribution has mean $\mu = 85$ and standard deviation $\sigma = 5$. A score is at $z = 0.5$. Find the raw score X .

$$\mu = 85, \quad \sigma = 5, \quad z = 0.5$$

→ Use the formula: $X = \mu + z * \sigma$.

$$\rightarrow X = 85 + (0.5)(5).$$

$$\rightarrow X = 87.5.$$

Answer: $X = 85 + (0.5)(5) = 87.5$

21. A distribution has mean $\mu = 69$ and standard deviation $\sigma = 14$. A score is at $z = 1.0$. Find the raw score X .

$$\mu = 69, \quad \sigma = 14, \quad z = 1.0$$

→ Use the formula: $X = \mu + z * \sigma$.

$$\rightarrow X = 69 + (1.0)(14).$$

$$\rightarrow X = 83.0.$$

Answer: $X = 69 + (1.0)(14) = 83.0$



22. A distribution has mean $\mu = 80$ and standard deviation $\sigma = 8$. A score is at $z = 1.5$. Find the raw score X .

$$\mu = 80, \quad \sigma = 8, \quad z = 1.5$$

→ Use the formula: $X = \mu + z * \sigma$.

$$\rightarrow X = 80 + (1.5)(8).$$

$$\rightarrow X = 92.0.$$

Answer: $X = 80 + (1.5)(8) = 92.0$

23. A distribution has mean $\mu = 97$ and standard deviation $\sigma = 19$. A score is at $z = 1.5$. Find the raw score X .

$$\mu = 97, \quad \sigma = 19, \quad z = 1.5$$

→ Use the formula: $X = \mu + z * \sigma$.

$$\rightarrow X = 97 + (1.5)(19).$$

$$\rightarrow X = 125.5.$$

Answer: $X = 97 + (1.5)(19) = 125.5$



Z-score from a normal distribution

24. A normal distribution has mean $\mu = 67$ and standard deviation $\sigma = 9$. Find the z-score for $x = 78$ and interpret it.

$$x = 78, \quad \mu = 67, \quad \sigma = 9$$

$$\rightarrow z = (x - \mu) / \sigma.$$

$$\rightarrow z = (78 - 67) / 9 = 11 / 9 = 1.22.$$

$\rightarrow z = 1.22$: score is 1.22 standard deviation(s) above the mean.

Answer: $z = \frac{78 - 67}{9} = 1.22$

25. A normal distribution has mean $\mu = 54$ and standard deviation $\sigma = 8$. Find the z-score for $x = 60$ and interpret it.

$$x = 60, \quad \mu = 54, \quad \sigma = 8$$

$$\rightarrow z = (x - \mu) / \sigma.$$

$$\rightarrow z = (60 - 54) / 8 = 6 / 8 = 0.75.$$

$\rightarrow z = 0.75$: score is 0.75 standard deviation(s) above the mean.

Answer: $z = \frac{60 - 54}{8} = 0.75$

26. A normal distribution has mean $\mu = 89$ and standard deviation $\sigma = 6$. Find the z-score for $x = 98$ and interpret it.

$$x = 98, \quad \mu = 89, \quad \sigma = 6$$

$$\rightarrow z = (x - \mu) / \sigma.$$

$$\rightarrow z = (98 - 89) / 6 = 9 / 6 = 1.5.$$

$\rightarrow z = 1.5$: score is 1.5 standard deviation(s) above the mean.

Answer: $z = \frac{98 - 89}{6} = 1.5$

27. A normal distribution has mean $\mu = 52$ and standard deviation $\sigma = 8$. Find the z-score for $x = 69$ and interpret it.

$$x = 69, \quad \mu = 52, \quad \sigma = 8$$

$$\rightarrow z = (x - \mu) / \sigma.$$

$$\rightarrow z = (69 - 52) / 8 = 17 / 8 = 2.12.$$

$\rightarrow z = 2.12$: score is 2.12 standard deviation(s) above the mean.

Answer: $z = \frac{69 - 52}{8} = 2.12$

28. A normal distribution has mean $\mu = 81$ and standard deviation $\sigma = 5$. Find the z-score for $x = 96$ and interpret it.

$$x = 96, \quad \mu = 81, \quad \sigma = 5$$

$$\rightarrow z = (x - \mu) / \sigma.$$

$$\rightarrow z = (96 - 81) / 5 = 15 / 5 = 3.0.$$

$\rightarrow z = 3.0$: score is 3.0 standard deviation(s) above the mean.

Answer: $z = \frac{96 - 81}{5} = 3.0$



29. A normal distribution has mean $\mu = 76$ and standard deviation $\sigma = 7$. Find the z-score for $x = 74$ and interpret it.

$$x = 74, \quad \mu = 76, \quad \sigma = 7$$

$$\rightarrow z = (x - \mu) / \sigma.$$

$$\rightarrow z = (74 - 76) / 7 = -2 / 7 = -0.29.$$

$\rightarrow z = -0.29$: score is 0.29 standard deviation(s) below the mean.

Answer: $z = \frac{74 - 76}{7} = -0.29$

30. A normal distribution has mean $\mu = 93$ and standard deviation $\sigma = 3$. Find the z-score for $x = 79$ and interpret it.

$$x = 79, \quad \mu = 93, \quad \sigma = 3$$

$$\rightarrow z = (x - \mu) / \sigma.$$

$$\rightarrow z = (79 - 93) / 3 = -14 / 3 = -4.67.$$

$\rightarrow z = -4.67$: score is 4.67 standard deviation(s) below the mean.

Answer: $z = \frac{79 - 93}{3} = -4.67$

